PROGRAM 12(a): Install MongoDB and Configure ATLAS

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AIM

To install MongoDB and configure MongoDB Atlas.

INTRODUCTION

MongoDB is a NoSQL, document-oriented database that stores data in JSON-like documents instead of traditional rows and columns.

MongoDB can be used in two ways:

1. Local Installation: Installing MongoDB software on your system.
2. Cloud-based Atlas: Using MongoDB Atlas, a fully managed cloud database service.

PART 1: INSTALL MONGODB LOCALLY (WINDOWS)

Step 1: Download MongoDB

• Go to MongoDB Community Server.

• Choose Windows.

• Select the MSI installer.

Step 2: Install MongoDB

1. Run the downloaded installer.
2. Select Complete Installation.
3. Make sure Install MongoDB as a Service is checked.
4. Select Run service as Network Service or Local System.
5. Check Install MongoDB Compass (GUI tool).
6. Click Next and Finish.

Step 3: Add MongoDB to PATH

1. Open Environment Variables.
2. Select System Variables → Path → Edit.
3. Click New.
4. Add the MongoDB bin path:

```
C:\Program Files\MongoDB\Server\6.0\bin
```

Note: The version number may differ.

Step 4: Verify installation

Open Command Prompt:

```
mongod --version
```

If installed correctly, it displays the MongoDB version.

PART 2: CONFIGURE MONGODB ATLAS (CLOUD)

Step 1: Create Atlas Account

• Go to MongoDB Atlas.

• Sign up using Google, GitHub or email.

Step 2: Create a Cluster

1. Click Build a Database.
2. Choose Free Tier (M0 – 512 MB storage, as mentioned in the record).
3. Select a cloud provider (AWS, Azure or GCP).
4. Select a region closest to you.
5. Click Create Cluster.

Step 3: Create a Database User

1. Go to Database Access.
2. Click Add New User.
3. Set username and password.
4. Save credentials.

Step 4: Configure Network Access

1. Go to Network Access.
2. Click Add IP Address.
3. Choose Allow Access from Anywhere (0.0.0.0/0) or add your local machine IP.

Step 5: Connect to Cluster

1. Click Connect.
2. Select Connect your application.
3. Get the connection string.

Example:

```
mongodb+srv://username:password@cluster0.abcd.mongodb.net/myDatabase
```

Step 6: Connect using MongoDB Shell

```
mongosh "mongodb+srv://cluster0.abcd.mongodb.net/myDatabase" --username myUser
```

BENEFITS OF MONGODB ATLAS

1. No need to install or maintain servers.
2. Provides automatic scaling and backups.
3. Accessible from anywhere with an internet connection.
4. Secure access control with authentication and IP whitelisting.

MONGODB ATLAS VS MONGODB COMPASS

MongoDB Atlas	MongoDB Compass
Hosts database in cloud	GUI tool to manage database
Handles server maintenance	Used for viewing and querying
Cloud service	Desktop application

PROGRAM 12(b): MongoDB CRUD Operations

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AIM

To write MongoDB queries to perform CRUD operations on documents using insert(), find(), update() and remove().

INTRODUCTION

CRUD stands for:

1. Create – Insert new documents.
2. Read – Retrieve documents.
3. Update – Modify existing documents.
4. Delete – Remove documents.

PROCEDURE

1. Open MongoDB Compass.
2. Click on Add New Connection.
3. Verify the MongoDB URL and port number.
4. Give a connection name and click Save & Connect.
5. Select a database.
6. Open MongoDB Command Shell.
7. Execute the following queries.

MONGODB QUERIES

1. Show Databases

```
show databases
```

Or

```
show dbs
```

2. Create / Switch Database

```
use school
```

Output: switched to db school

3. Create Collection

```
db.createCollection("students")
```

4. CREATE – Insert Documents

Insert a single document:

```
db.students.insert({
    rollno: 1,
    name: "Ravi",
    department: "CSE",
    age: 20
})
```

Insert multiple documents:

```
db.students.insert([
    {rollno: 2, name: "Anita", department: "CSE", age: 21},
    {rollno: 3, name: "Kiran", department: "ECE", age: 20},
    {rollno: 4, name: "Sita", department: "CSE", age: 19}
])
```

5. READ – Find Documents

Display all documents:

```
db.students.find()
```

Find a specific document:

```
db.students.find({department: "CSE"})
```

6. UPDATE – Update Documents

Update one document:

```
db.students.update(
    {rollno: 1},
    {$set: {age: 22}}
)
```

7. DELETE – Remove Documents

Remove one document:

```
db.students.remove({rollno: 1})
```

Remove all documents:

```
db.students.remove({})
```

OUTPUT

The MongoDB queries are executed successfully to insert, retrieve, update and delete documents in the students collection.

RESULT

Thus, MongoDB CRUD operations using insert(), find(), update() and remove() were performed successfully.

IMPORTANT VIVA QUESTIONS

1. What is CRUD?

CRUD stands for Create, Read, Update and Delete.